AN INDIAN ADAPTATION

Third Edition

Supply Chain Management

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SUBJECT: SUPPLY CHAIN MANAGEMENT STRATEGY

CHAPTER: 06

CHAPTER TITLE: SOURCING

REFERENCE BOOK:

SUPPLY CHAIN MANAGEMENT (AN INDIAN ADAPTATION), N R SANDERS & S S PADHI, WILEY, 3RD ED. (2023)

KEY CONCEPTS

Sourcing	*The business function responsible for **all activities and processes required to purchase goods and services from suppliers**.*
Purchasing	*The process of **buying goods and services**. It includes supplier identification and selection, buying, negotiating contracts, and measuring supplier performance.*
Strategic Sourcing	*This requires expanding the role of sourcing from mere buying, **to building close and longer-term working relationships with selected suppliers and partners**.*
Maverick Buying	*A practice where internal users attempt to buy directly from suppliers and bypass the sourcing function.*
Fair Price	*The **lowest price** that can be paid while ensuring a continuous supply of quality goods.*
Total Cost of Ownership	***The purchase price plus all other costs** associated with acquiring the item.*
Competitive Bidding	*One of the methods to reach agreement and develop contracts with potential suppliers. This has the objective of awarding the business to the most qualified bidder, once specific criteria have been identified.*
Negotiation	*It is **a communication process between two parties** that attempts to reach a mutual agreement.*

KEY CONCEPTS

Functional Products	*Functional products are those that **satisfy basic functions or needs.** They include items such as **food** purchased in grocery stores, **gas and oil** for transportation and heating.*
Innovative products	*The products are purchased for other reasons, such as **innovation and status, and include high fashion items or technology products**, such as those seen in the computer industry.*
Stable Supply Process	*A stable supply process is where sources of supply are **well established, manufacturing process used are mature, and the underlying technology is stable.***
Evolving Supply Process	*An evolving supply process is where sources of supply are **rapidly changing, the manufacturing process is in an early stage, and the underlying technology is quickly evolving.***
Postponement	*The customization strategy of supply chain **to delay product differentiation as late in the supply chain as possible.***
Off-shoring	***The use of global sourcing** to reduce costs and access cheaper labor.*
Re-shoring	***Returning to the domestic sources.***

KEY CONCEPTS

Outsourcing	*Choosing a third party, such as an outside supplier to perform a task to incur business benefits.*
Scope	***The degree of responsibility*** *assigned to the suppliers.*
Criticality	*The* ***importance of the outsourced activities or tasks*** *to the organization industry.*
E-auctions	*E-auctions are the* ***use of internet technology to conduct auctions*** *as a means of selecting suppliers.*

SECTION-1

What Is Sourcing?

- Sourcing is the business **function responsible for all activities and processes required to purchase goods and services from suppliers**.

- The sourcing function has the primary responsibility for the supply side of the organization.

- The sourcing addresses the ***upstream*** part of the supply chain.

- Figure 6.1 shows the functional position of sourcing within the organizational framework.

- The sourcing function is often referred to by a number of different terms, and there is some confusion regarding its name. These names include ***purchasing, procurement, sourcing, strategic sourcing and supply management.***

- Although they are used interchangeably, they do not necessarily mean the same thing.

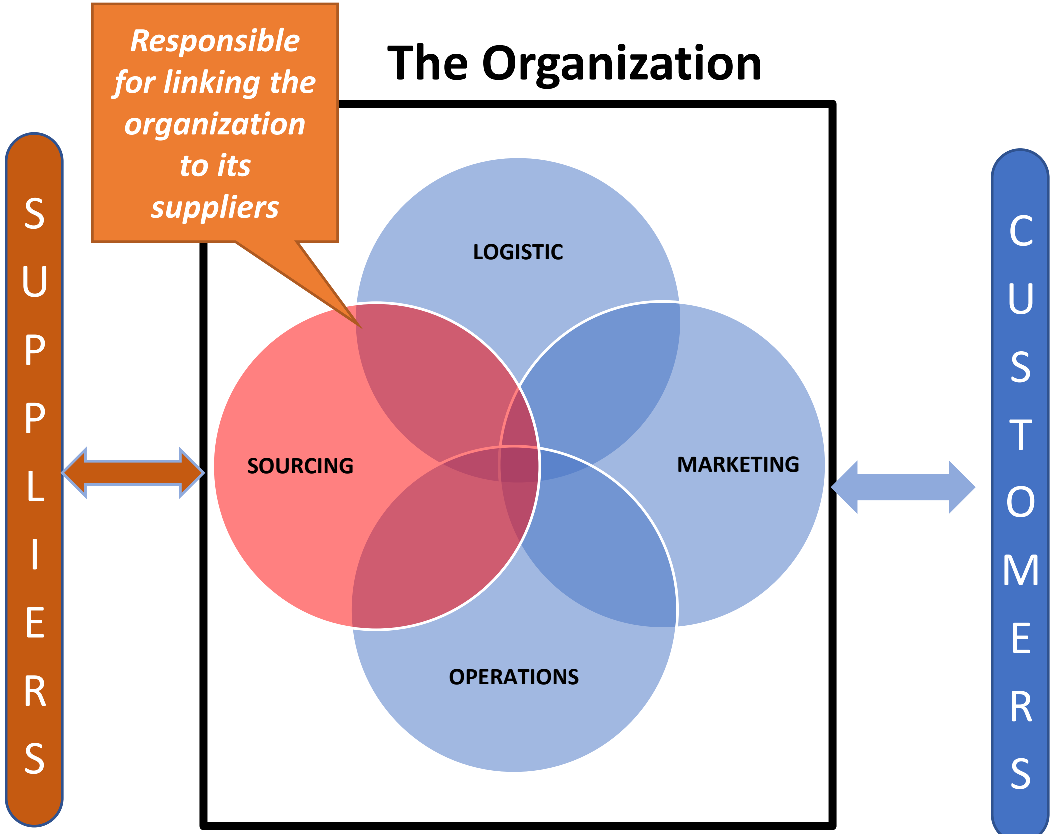

Fig. 6.1 The Sourcing Function

✓ Here sourcing is used to refer to the overall management of the supplier base, encompassing all the terms and range of activities, from tactical to strategic.

Example of Sourcing

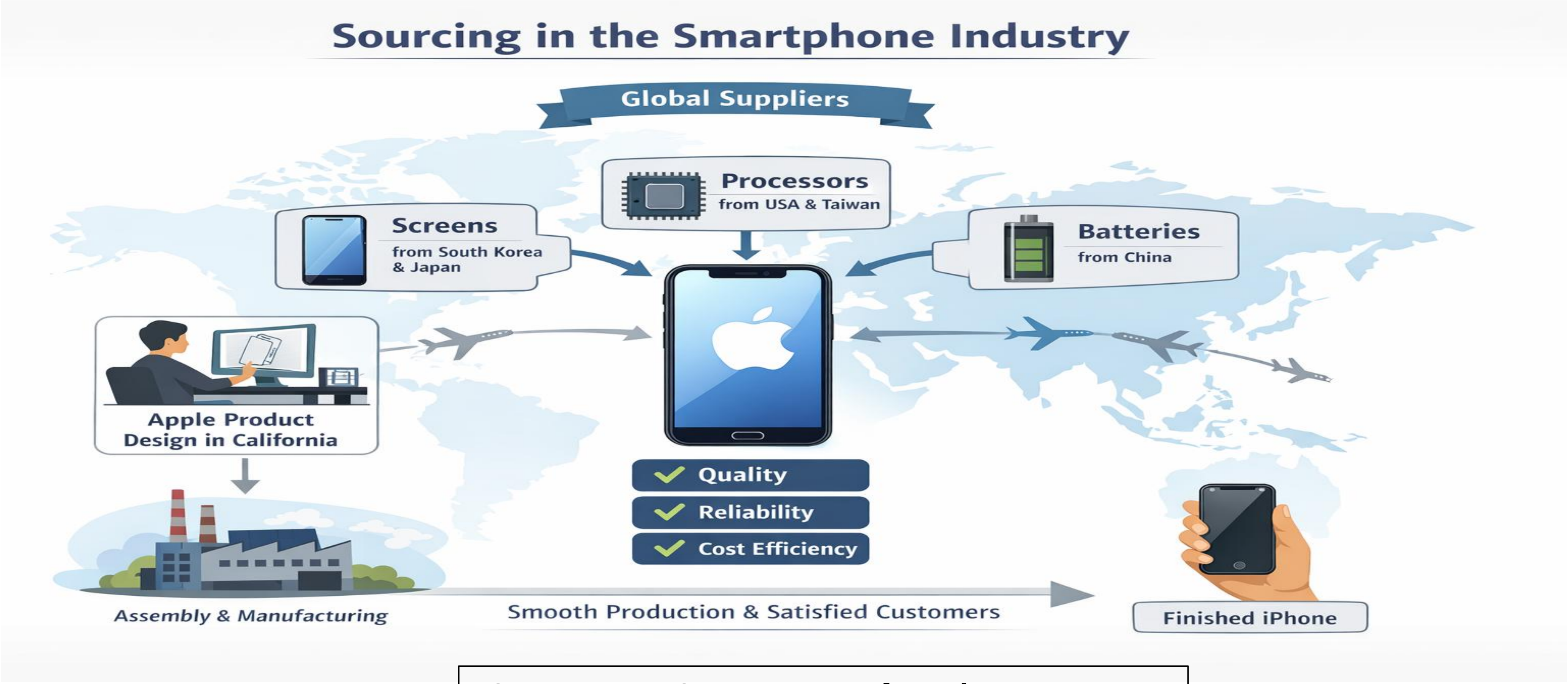

Fig. 6.2 Sourcing process of Apple

A good **example of sourcing** can be seen in the smartphone industry, as shown in the figure. 6.2. Companies like Apple design their products but rely on multiple suppliers around the world to provide components such as processors, screens, and batteries. The company carefully selects suppliers based on quality, reliability, and cost efficiency to ensure that production runs smoothly and the final product meets customer expectations.

Example of Sourcing

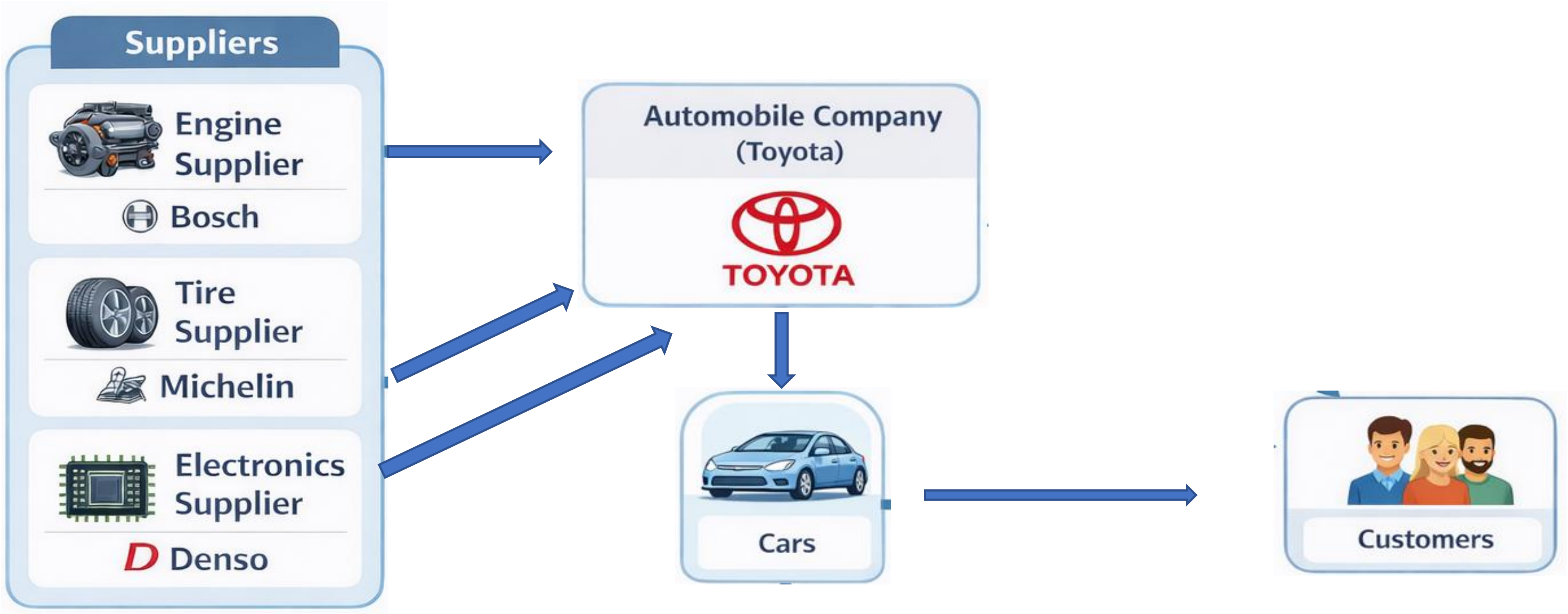

Fig. 6.3 Sourcing process of Toyota

Another **example of sourcing** can be observed in the automobile industry, as shown in the figure. 6.3. Companies like Toyota design and assemble vehicles, but they rely on a large network of suppliers to provide critical components such as engines, tires, and electronics. These automobile manufacturers carefully evaluate suppliers based on factors such as product quality, delivery reliability, technological capability, and cost efficiency. By sourcing components from specialized suppliers, the company ensures efficient production and maintains high product quality.

Purchasing Vs Strategic Sourcing

- ***Purchasing*** is the process of buying goods and services.
- It is a narrow functional activity that include ***supplier identification and selection, buying negotiating contracts, and measuring supplier performance***.
- ***Strategic sourcing*** goes beyond focusing on just the price of goods to looking at the strategic function from a strategic and future-oriented perspective.
- Strategic sourcing is not the same as purchasing and involves a much more progressive approach to the sourcing function.
- This requires expanding the role of sourcing from mere buying to building a longer-term working relationship with specially selected suppliers and partners.
- Therefore, ***purchasing and strategic sourcing are not the same***. As shown in fig. 6.4, purchasing mainly involves tactical activities related to routine buying, whereas strategic sourcing involves broader strategic planning and long-term supplier relationships.

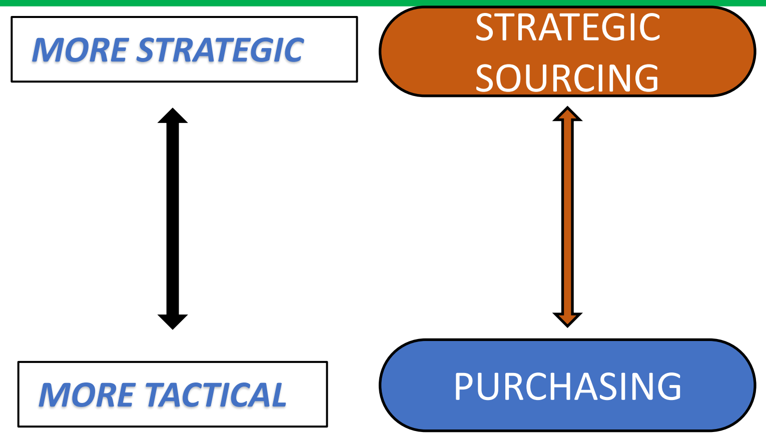

Fig. 6.4 Hierarchy of terms

- ❖ A small retail shop purchasing office stationery from a local vendor is an example of **purchasing**, because the transaction is simple, short-term, and primarily focused on obtaining items at a reasonable price.
- ❖ In contrast, a large automobile manufacturer such as Toyota engaging in long-term partnerships with component suppliers represents ***strategic sourcing***, as the company works closely with its suppliers to improve product quality, reduce costs, and ensure reliable supply over time.

Commercial Vs Consumer Sourcing

- ***Commercial sourcing*** *refers to* ***businesses purchasing goods or services to use in production, operations, or resale.***

- ***Consumer sourcing or buying*** refers to ***individual customers buying goods or services for personal use***.

- *Commercial sourcing* is different from *consumer buying*. One difference relates to the ***number of buyers versus suppliers***.
- **In consumer buying or sourcing**
  - There are many suppliers of common items, and the buyers are typically final consumers of the items they purchase.

  - Each buyer comprises a relatively small portion of the supplier's total business. Hence, little ability to negotiate the purchase price.

  - ***Example-*** *A consumer purchasing a laptop typically compares brands such as Dell, HP, or Lenovo by considering factors like price, product features, reviews, and brand reputation. The purchase quantity is small (usually one unit), and the decision is based mainly on personal preferences and budget rather than long-term supply contracts.*

Commercial Vs Consumer Sourcing

Commercial sourcing

- The main function is acquiring the right material and services and making sure they are available to all parts of the organization in the right quantity, at the right price, at the right time.
- It includes everything from materials directly used in manufacturing to support services and office supplies.
- The volumes purchased are on a large scale and organizations typically have very specialized purchasing needs.
- Commercial sourcing must also make sure that it follows all legal and ethical procedures and abides by environmental and security regulations.
- ***Example-*** *A car manufacturer such as Toyota sources steel, tires, electronic components, and engines from multiple suppliers. The sourcing department evaluates suppliers based on price, quality, reliability, delivery capabilities, and potential for a long-term partnership. Large purchase volumes and formal contracts are involved. The objective is to ensure a continuous supply of materials needed for production while minimizing cost and maintaining quality.*

Commercial Vs Consumer Sourcing

Commercial Sourcing	*Consumer Sourcing*
*Buyers are **organizations or businesses** purchasing goods for production or operations.*	*Buyers are typically **final consumers** who purchase products for personal use.*
*The **number of potential suppliers may be limited** due to specialized requirements.*	*There are **many suppliers** available for common consumer products.*
*Buyers often represent a **large portion of the supplier's business** because of high purchase volumes.*	*Each individual buyer represents **only a small portion of the supplier's total business**.*
*Organizations often have **significant negotiating power** because they purchase in large quantities.*	*Individual consumers generally have **little ability to negotiate prices**.*
*Purchases involve **large volumes and large sums of money**.*	*Purchase transactions are usually **small in volume and relatively simple**.*
*Buying decisions are **more complex and structured**.*	*Buying decisions are generally **informal and less complex**.*
*Relationships often involve **formal contracts and long-term agreements**.*	*Relationships between buyer and seller are usually **short-term or transactional**.*

Impact on the Organization and Supply Chain

Operational Impact

Operational impact refers to the role sourcing plays in ensuring the smooth functioning of organizational activities and supply chain operations.

The performance of an organization and its supply chain depends heavily on an **efficient sourcing function** that ensures the right materials are available throughout the organization in the **right quantities,** at the **right places,** at the **right time.**

Sourcing managers play a critical role in maintaining the ***balance between too little and too much inventory***.

•If there are **too few materials**, the organization may not be able to function because production activities will stop.

•If there is **too much inventory**, it leads to **tied-up capital and financial losses**.

Impact on the Organization and Supply Chain

Financial Impact

- Sourcing also has a **critical financial impact on the organization** because companies spend a large amount of money on purchasing goods and services from suppliers.
- Proper management of sourcing activities can result in **significant cost savings**, which can greatly affect organizational profitability.
- In most manufacturing organizations, sourcing represents the **largest category of company spending**, which may range from **50% to 90% of the organization's revenue**.
- For example, in the automobile industry, almost **80% of the total cost of an automobile comes from purchased components**, while the manufacturing facility mainly performs the assembly of these purchased parts.

Impact on the Organization and Supply Chain

Strategic Impact

- All sourcing decisions must support the **overall business strategy of the organization**. Therefore, sourcing managers must select suppliers that help the organization achieve its **competitive priorities**.

  - Companies that compete primarily on **cost** will focus on selecting suppliers that can provide materials at the lowest possible cost.

  - Companies that compete on **quality or innovation** will focus on suppliers that can provide superior quality materials or advanced technologies.

- Poor supplier quality can lead to: increased **scrap,** higher number of **returned products**, **slower manufacturing processes**

- On the other hand, poor supplier management may result in the organization **paying higher prices for supplies compared to competitors**.

Impact on the Organization and Supply Chain

Risk Mitigation Impact

- The sourcing function helps organizations **reduce the risk of supply disruptions** and protect the company's reputation.
- Sourcing is **not a one-time activity**, but a **continuous process**, and organizations must ensure a **reliable and continuous supply of materials**.
- For this reason, suppliers must be evaluated not only for immediate purchases but also as **long-term supply partners**.
- In addition, sourcing managers must ensure that suppliers meet several **performance standards**, including legal standards and ethical standards.
- Sourcing from suppliers that behave **unethically**, even if their prices are low, can **damage the company's image and reputation**.

Impact on the Organization and Supply Chain

Information Impact

The sourcing function continuously collects and analyzes information related to:

- ✓ Availability of suppliers
- ✓ Prices of materials
- ✓ New products
- ✓ Emerging technologies

This information is valuable because it brings **new knowledge into the organization and its supply chain**.

Such knowledge helps organizations in many ways, including:

- ✓ Understanding the **true cost of products**
- ✓ Identifying **new technologies**
- ✓ Supporting **product design and innovation**

Sourcing Issues

Managing Customer Expectations

- Customers today expect products that are **high in quality, competitively priced, and delivered quickly**.
- Because sourcing decisions directly influence product cost, quality, and delivery performance, sourcing managers must ensure that suppliers are capable of meeting these expectations.
- If suppliers fail to meet required standards, it may lead to **production delays, higher costs, and dissatisfied customers**, which ultimately affects the organization's performance.

Managing Suppliers

- Organizations must develop **strong and effective relationships with reliable suppliers** to ensure consistent supply and quality.
- Long-term collaboration with suppliers can help organizations achieve **better coordination, improved communication, and greater supply stability**.

Maintaining quality and Sustainability

- Companies are increasingly expected to ensure that their suppliers comply with **environmental regulations, labor standards, and ethical business practices**.
- Selecting suppliers that follow responsible practices helps organizations maintain a positive reputation and avoid legal or ethical problems.

Access to Accurate Information

- Effective sourcing decisions require access to **reliable and timely information** about suppliers, prices, product specifications, and technological developments.
- Without accurate information, organizations may make poor sourcing decisions that increase costs or reduce efficiency.

Sourcing managers must identify potential risks and develop strategies to ensure a continuous and reliable supply of materials and services.

SECTION-2

THE SOURCING FUNCTION

Sourcing Function

Sourcing Process

1. Identification: The process begins when an organization recognizes a requirement for a product or service. This need may arise from production departments, operations teams, or other functional areas that require materials to support their activities.

2. Specification Development: After identifying the need, the organization prepares detailed specifications describing the product or service required. These specifications outline aspects such as quality standards, technical characteristics, and quantity requirements.

3. Supplier Search: At this stage, the organization identifies potential suppliers capable of providing the required goods or services. Suppliers may be located through market research, supplier databases, industry contacts, or previous sourcing relationships.

4. Request for Information / Quotation / Proposal: The organization communicates with potential suppliers by requesting information, quotations, or proposals. Suppliers provide details about their capabilities, pricing, delivery terms, and other relevant conditions.

5. Supplier Evaluation and Selection: The organization evaluates supplier responses based on criteria such as cost, quality, delivery reliability, and overall capability. After comparing alternatives, the organization selects the supplier that best meets its requirements.

6. Contract Negotiation: Once a supplier is selected, the organization negotiates terms and conditions, including pricing, delivery schedules, payment arrangements, and service expectations. These terms are documented in a formal contract.

7. Supplier Performance Monitoring: The sourcing process continues even after the contract is signed. Organizations regularly monitor supplier performance to ensure suppliers meet expectations for quality, delivery, and reliability.

Fig. 6.5 sourcing Process

Cost Vs Price

Price: The amount at which an item is sold in the marketplace. It must cover the supplier's total cost plus a profit margin.

Cost: The sum of all expenses incurred to produce a product, categorized into ***fixed*** *(costs that the company incurs regardless of how much it produces) and variable costs (costs that vary directly with the amount of units produced).*

Example- Fixed cost: ***overhead, taxes and insurance***
Variable cost: ***direct materials and labor***

$$\text{Total Cost} = F + V_c Q$$

Where, *F= Fixed cost*
V_c= Variable cost per unit
Q = number of units sold

The goal of developing purchasing contracts and negotiation is to find a ***fair price,*** which is the lowest price that can be paid, ensuring a continuous supply of quality goods. This price must cover the total cost and provide a profit to keep the business.

Total cost of ownership (TCO): *It is the purchase (selling) price plus all other costs associated with acquiring the item (i.e.,* ***transportation, administrative costs, follow-up, storage, inspection and testing****). This helps to evaluate the price in purchasing before selecting a supplier.*

Variable cost	
Materials	$ 5,000
Labor	$ 2,000
+	
Fixed Cost	
Facility Overhead	$ 3,500
Total Cost	$ 10,500
+	
Profit	$ 1,000
Selling Price	$ 11,500
+	
Transportation cost	$ 1000
Total cost of ownership (TCO)	$ 12,500

Bidding or Negotiation

Competitive Bidding	***Negotiation***
One of the methods to reach agreement and develop contracts with potential suppliers. This has the objective of awarding the business to the most qualified bidder, once specific criteria have been identified.	*It is a communication process between two parties that attempts to reach a mutual agreement.*
Performance criteria and specifications can be clearly defined	*Performance criteria and specifications cannot be clearly defined*
Volume purchased is high	*Many criteria exist for supplier selection (e.g. cost, quality, delivery, and service)*
Products being purchased are standardized	*Product being purchased is new or technically complex*
Many qualified suppliers	*Few qualified suppliers exist in the marketplace*
Time is available for bid evaluation	*Suppliers customize the product to buy*

SECTION-3
Sourcing and SCM

Functional Vs Innovative Products

- *Demand and supply uncertainties are related to the type of products and their supply chains. Based on this, the products can be classified as functional and innovative.*

- *Functional products are those that **satisfy basic functions or needs.** They include items such as **food** purchased at grocery stores, **gas and oil** for transportation and heating.*

Characteristics of Functional Products

Functional products generally have the following characteristics:

- ✓ **Stable and predictable demand :**
- ✓ **Long product life cycles**
- ✓ **Low profit margins**
- ✓ **High competition among suppliers**
- ✓ **Little product variety**
- ✓ **Efficient and cost-focused supply chains**

- *These products are usually purchased frequently, and customers often choose them based on **price and availability rather than uniqueness**.*

Examples of Functional Products

Typical examples of functional products include:

- ✓ Basic household items
- ✓ Food products such as **flour, sugar, and rice**
- ✓ Standard office supplies
- ✓ Basic clothing items

- *Because these products are widely available and demand is predictable, organizations focus on **cost efficiency and operational efficiency** in managing their supply chains.*

Functional Vs Innovative Products

- *The products are purchased for other reasons, such as* ***innovation and status, and include high-fashion items or technology products****, such as those seen in the computer industry.*

Characteristics of Innovative Products

Innovative products typically exhibit the following characteristics:

- ✓ Unpredictable demand
- ✓ Short product life cycles
- ✓ Higher profit margins
- ✓ Frequent product design changes
- ✓ Higher risk of stockouts or excess inventory

- These products require supply chains that can **respond quickly to changes in market demand**.

Examples of innovative products include:

- ✓ New electronic gadgets
- ✓ Fashion apparel with rapidly changing styles
- ✓ Newly launched technology products

- These products often rely on **responsive supply chains** that emphasize speed, flexibility, and rapid product development.

Stable Vs Evolving Supply Process

- Supply processes can generally be categorized into **stable supply processes** and **evolving supply processes**. This classification depends on the **level of uncertainty associated with the supply of products and materials**.
- A ***stable supply process*** refers to a situation where the supply of a product is **well understood, predictable, and consistent**. Production technologies are mature, and suppliers are capable of delivering products reliably.

Characteristics of Stable Supply Processes

Stable supply processes generally have the following characteristics:

- ✓ **Established and proven production technologies**
- ✓ **Predictable production yields**
- ✓ **Reliable suppliers**
- ✓ **Low supply uncertainty**
- ✓ **Well-developed supply networks**

- Because the supply conditions are predictable, organizations can focus on **cost efficiency and streamlined operations**.

Examples of Stable Supply Processes

Examples include production and supply of:

- ✓ Basic raw materials
- ✓ Standard industrial components
- ✓ Common consumer goods with established production methods

- In such cases, suppliers have extensive experience and the production processes are **well standardized**.

Stable Vs Evolving Supply Process

- An ***evolving supply process*** occurs when the production or supply of a product involves **new technologies, uncertain production processes, or limited supplier capabilities**.
- Because the process is still developing, supply conditions may be **uncertain and difficult to predict**.

Characteristics of Evolving Supply Processes

Evolving supply processes typically involve:

- ✓ **New or developing technologies**
- ✓ **Uncertain production yields**
- ✓ **Limited number of capable suppliers**
- ✓ **Higher supply uncertainty**
- ✓ **Frequent changes in production processes**

- Organizations dealing with evolving supply processes must focus on **flexibility, collaboration with suppliers, and risk management**.

Examples of Evolving Supply Processes

Examples include supply of:

- ✓ Newly developed technological products
- ✓ Advanced electronic components
- ✓ Products using emerging manufacturing technologies

•In these cases, suppliers may still be **refining their production methods**, leading to greater uncertainty.

Demand vs Supply Uncertainty Framework

•**Demand Uncertainty** refers to how predictable customer demand is.
•**Supply Uncertainty** refers to how predictable the supply process is.
Combining these two dimensions helps organizations determine the **appropriate sourcing and supply chain strategy**.

The Four Supply Chain Strategies

1. Efficiency-focused Supply Chain

Used when **demand uncertainty is low and supply uncertainty is low**.
- ✓ Focus on **cost efficiency and operational excellence**.
- ✓ Suitable for **functional products with stable supply processes**.

2. Risk-Hedging Supply Chain

Used when **demand uncertainty is low but supply uncertainty is high**.
- ✓ Organizations pool and share resources to **reduce supply risks**.
- ✓ Emphasis on **supply reliability**.

3. Responsive Supply Chain

Used when **demand uncertainty is high but supply uncertainty is low**.
- ✓ Focus on **responding quickly to changing customer demand**.
- ✓ Suitable for **innovative products with predictable supply processes**.

4. Agile Supply Chain

Used when **both demand and supply uncertainty are high**.
- ✓ Requires **high flexibility and adaptability**.
- ✓ Organizations must respond quickly to **both demand fluctuations and supply disruptions**.

		Demand uncertainty	
		Low (functional products)	High (innovative products)
Supply uncertainty	Low (stable process)	Efficiency-focused SC	Responsive SC
	High (evolving process)	Risk-hedging SC	Agile SC

Fig. 6.6 Demand vs Supply Uncertainty

Single Vs Multiple Sourcing

Single Sourcing

Single sourcing refers to the practice of purchasing a particular product or service from **only one supplier**. The organization relies on a single supplier to meet its requirements for that specific item.

Advantages of Single Sourcing

Single sourcing offers several potential benefits:

- ✓ **Stronger buyer–supplier relationships**
- ✓ **Better communication and coordination with the supplier**
- ✓ **Possibility of quantity discounts due to larger purchase volumes**
- ✓ **Improved product quality through long-term collaboration**
- ✓ **Reduced administrative and transaction costs**

Disadvantages of Single Sourcing

Despite its advantages, single sourcing also involves certain risks:

- ✓ **High dependency on a single supplier**
- ✓ **Risk of supply disruption if the supplier fails**
- ✓ **Limited bargaining power for the buyer**
- ✓ **Potential problems if the supplier experiences financial or operational difficulties**

Single Vs Multiple Sourcing

Multiple Sourcing

Multiple sourcing refers to the practice of purchasing a product or service from **two or more suppliers**.

Instead of relying on a single supplier, organizations distribute their purchases among several suppliers.

Advantages of Multiple Sourcing

Multiple sourcing provides several benefits:

- ✓ **Reduced risk of supply disruptions**
- ✓ **Greater flexibility in sourcing decisions**
- ✓ **Increased competition among suppliers**
- ✓ **Better bargaining power for the buyer**

Disadvantages of Multiple Sourcing

However, multiple sourcing also has certain limitations:

- ✓ **Higher administrative and coordination costs**
- ✓ **Smaller order quantities per supplier**
- ✓ **Weaker relationships with individual suppliers**
- ✓ **Potential difficulties in maintaining consistent quality across suppliers**

Managing several suppliers may require **additional effort in coordination, monitoring, and quality control**.

Domestic Vs Global Sourcing

Domestic Sourcing

- ✓ Procurement of goods or services from **suppliers located within the same country**.
- ✓ Helps **reduce transportation costs**, allows firms to **monitor quality closely**, and maintain **closer supplier relationships**.
- ✓ Recently, many firms are returning to domestic sourcing to improve **control and responsiveness**.

Global Sourcing

- ✓ Procurement of goods or services from **suppliers located in other countries**.
- ✓ Has increased as firms seek **lower labor costs and specialized capabilities** in different regions.
- ✓ Used in industries such as **manufacturing, service operations, and software development**.

Offshoring

- ✓ A form of global sourcing where companies **move production or service activities to another country** to benefit from **lower labor costs**.
- ✓ Examples include **data processing, software development, and diagnostic test analysis**.

Reshoring

- ✓ The process of **bringing previously offshored activities back to the home country**.
- ✓ Companies re-shore to **reduce transportation costs, improve quality monitoring, and strengthen supplier relationships**.

Outsourcing

- ***Outsourcing*** *refers to the practice of* ***hiring an outside company to perform tasks, provide services, or manufacture goods that were traditionally performed within the organization****.*

- *Companies outsource activities to* ***specialized external providers*** *in order to improve efficiency and reduce*

Outsourcing decisions can be analyzed using **two key dimensions**:

1. Scope

Scope refers to **how much of the activity or process is outsourced**.

- ✓ **Low Scope** → Only a **small portion of the activity** is outsourced.
- ✓ **High Scope** → A **large portion or entire function** is outsourced to external suppliers.

2. Criticality

Criticality refers to **how important the outsourced activity is to the organization's operations or competitive advantage**.

- ✓ **Low Criticality** → The activity is **not central to the firm's core operations**.
- ✓ **High Criticality** → The activity is **strategically important and may require closer monitoring**.

- A common example of outsourcing is the use of **third-party logistics providers (3PLs)**.
- Companies outsource **transportation and logistics activities** to specialized providers such as: **UPS**, **FedEx**
- These companies provide **logistics and delivery services**, allowing firms to **avoid maintaining their own transportation systems and logistics infrastructure**.

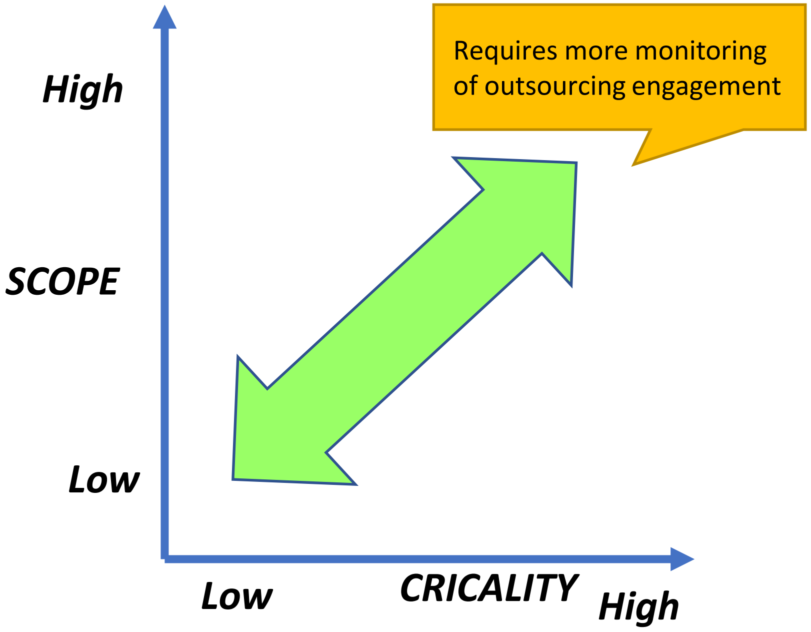

Fig. 6.7 Dimensions of outsourcing

Electronic Auctions (E-Auctions)

- **E-auctions** are the use of Internet technology to conduct auctions as a means of selecting suppliers and determining aspects of the purchase contract.

- Through e-auctions, buyers can determine: **Price, Product quality**, **Volume allocation among suppliers**, **Delivery schedules**

Benefits of E-Auctions

- ✓ **Market transparency**
- ✓ **Decreased error rate**
- ✓ **Simplified comparison of sources of supply**
- ✓ **Increased buying reach** because companies can access a broader supplier base through the Internet
- ✓ **Reduction in ordering cycle time** since the process is automated and more efficient than traditional sourcing methods such as **RFP or RFQ**

Potential Problems with E-Auctions

Despite the benefits, there are some concerns:

- ✓ Suppliers may **bid unrealistically low prices** in order to win the auction and later attempt to renegotiate.
- ✓ More realistic suppliers may **drop out early in the bidding process**.
- ✓ Some suppliers may participate **only to gather market intelligence**.
- ✓ E-auctions can **damage long-term supplier relationships** because existing suppliers may feel pressured by the competitive bidding environment.

Electronic Auctions (E-Auctions)

Types of E-Auctions

E-auctions can differ based on **how competition among buyers and sellers occurs**.
Common types include:

Open Auction

- ✓ Suppliers **can see the most competitive bids** during the auction.
- ✓ Suppliers can **adjust their bids until the auction closes**.

Sealed-Bid Auction

- ✓ Suppliers **submit bids without seeing competitors' bids**.
- ✓ The buyer evaluates all bids **after submission**.

Reverse Auction

A reverse auction is a type of e-auction in which:

- ✓ Multiple suppliers compete to win a buyer's contract
- ✓ Suppliers continuously lower their bids in order to offer the lowest price

In contrast to traditional auctions (where buyers bid higher prices), in a reverse auction:

- ✓ The price decreases as suppliers compete
- ✓ This process helps buyers obtain more competitive pricing from suppliers.

Electronic Auctions (E-Auctions)

Conducting an E-Auction

Before conducting an e-auction, organizations must:

✓ Clearly specify **product requirements** such as quality, quantity, delivery, service, and contract length.

✓ **Train participants** on the auction technology.

✓ **Test the auction system** before the event.

The buyer must also define **rules of the auction**, such as:

✓ Duration of the auction

✓ Whether **closing time can be extended**

•How suppliers will be **notified of bid rankings**

SECTION-4

Measuring Sourcing Performance

Kraljic Purchasing Matrix (KPM)

Kraljic Purchasing Matrix (KPM)

- The **Kraljic Matrix** is a **strategic tool developed by Peter Kraljic in 1983** to help purchasing managers **analyze purchasing strategies and manage supplier relationships effectively**.
- The role of the **Kraljic Purchasing Matrix (KPM)** becomes more important as the **value and importance of the purchasing function increase**.

A fundamental aspect of **supply chain management** is the **sourcing and negotiation of supplies** to support business operations.

Purchasing contracts usually include terms such as:

- ✓ **Product specifications**
- ✓ **Price paid**
- ✓ **Quantity discounts**
- ✓ **Information sharing**

Strategic sourcing often uses analytical models like the **Kraljic Matrix** to help firms **analyze their list of purchases and determine the importance of each item**.

Kraljic Purchasing Matrix (KPM)

Kraljic Purchasing Matrix (KPM)

The Kraljic matrix classifies products and services using two key dimensions:

Supply Risk

Supply risk refers to the uncertainty related to the **availability and reliability of suppliers**.

Supply risk can arise due to several factors:

1.Market Risk

1. Availability of potential suppliers
2. Market structure such as **monopoly or oligopoly**
3. Entry barriers to the market

2.Performance Risk

1. Supplier quality and performance issues
2. Supplier's **access to and adoption of new technology**

3.Complexity Risk

1. Problems associated with **standardization**
2. When **product or service specifications are critical**

Purchasing Impact (Impact on the Bottom Line)

This dimension measures the **importance of the purchased item to the firm's profitability and operations**.

Factors considered include:

1.Impact on Profitability: The effect of purchasing decisions on the **profit yield of the firm**.

2.Importance of Purchase: The **importance or necessity of the purchased item** for the organization.

3.Value/Cost of Purchase: The **tangible and intangible costs** associated with the purchase and the **value obtained from it**.

Purchasing Portfolios Based on KPM

Purchasing's impact on the bottom line	Low supply risk	High supply risk
High	**Leverage products** ▪ Alternative sources of supply available ▪ Substitution possible ↓ **Competitive bidding**	**Strategic products** ▪ Critical for product's cost price ▪ Dependence on supplier ↓ **Performance based partnership**
Low	**Routine products** ▪ Large product variety ▪ High logistics complexity ▪ Labor intensive ↓ **Category management and e-procurement**	**Bottleneck products** ▪ Monopolistic market ▪ Large entry barriers ↓ **Secure supply**

Fig. 6.8 Purchasing portfolios for different types of products

Purchasing's impact on the bottom line	*Low* — *Supply risk*	*High*
High	**Leverage products** ▪ Utilize synergies within the group ▪ Optimize costs, conditions ▪ International procurement ▪ Framework agreements	**Strategic products** ▪ Long-term partnerships ▪ Single sourcing with key suppliers ▪ Joint development ▪ Integrated processes ▪ Supplier management ▪ Collaboration agreements
Low	**Routine products** ▪ Simplification process ▪ Single sourcing ▪ Supplier reduction ▪ Defined ranges, utilize standardization within the firm	**Bottleneck products** ▪ Value analysis and standardization ▪ Identify alternative suppliers (procurement marketing) ▪ Establish alternative suppliers (suppliers development)

Measuring Sourcing Performance

- There are two common performance measures that can be used to measure the performance of the sourcing function relative to its utilization of inventory, i.e., ***inventory turnover and weeks-of supply*.**

Inventory Turnover

- It measures how quickly inventory moves. It is computed as:

$$\text{Inventory turnover} = \frac{\textit{Cost of goods sold}}{\textit{Average Inventory Value}}$$

- In general, the higher the inventory turnover rate the better the company utilizes its inventory. However, some companies need to turn over faster than others. Ex- *perishable items such as food or medicine, innovative products with short life cycles.*

Example-

- Jenco incorporated is a toy producer, experiencing a high inventory cost and is concerned that its inventory is selling at an appropriate rate. It has an annual cost of goods sold of \$8,000,000 and an average inventory is \$2,000,000. What is the annual turnover?

$$\text{Inventory turnover} = \frac{\$8{,}000{,}000}{\$2{,}000{,}000} = 4 \text{ inventory turns per year}$$

Measuring Sourcing Performance

Weeks-of-Supply

- It provides the length of time demand can be met with on hand inventory. It is computed as:

$$\text{weeks of supply} = \frac{\textbf{Average on hand inventory}}{\textbf{Average weekly usage}}$$

- It tells the manager how long the current on-hand inventory will last based on current demand. A large value means too much inventory is being held, whereas, a value that is too low increases the chances of stock-out or lost sales.

Example-

- Jenco incorporated is concerned about having enough inventory in stock and wants to compute its weeks of supply with the given annual cost of goods sold of $8,000,000 and inventory $2,000,000.

Solution: weekly cost of goods sold can be computed by dividing the annual cost of goods sold by 52 weeks per year. Weeks of supply can then be computed as:

$$weeks\ of\ supply = \frac{\$2{,}000{,}000}{\$8{,}000{,}000/52}$$

$$= \frac{\$2{,}000{,}000}{\$153{,}846.15} = 13\ \textit{weeks of supply}$$

Practice Questions

Practice Set-

- A supplier produces a component with the following cost structure:
  - **Fixed cost (F)** = $120,000
  - **Variable cost per unit (Vc)** = $20
  - **Expected production quantity (Q)** = 5,000 units
  - **Supplier's profit margin** = $5 per unit

  Additional buyer-related costs are:
  - Transportation = $8,000
  - Administrative cost = $2,000
  - Inspection and testing = $3,000
  - Storage cost = $4,000

  Questions:
  1. Calculate the **purchase price per unit** offered by the supplier.
  2. Calculate the **total purchase cost for all units**.
  3. Determine the **Total Cost of Ownership (TCO)** for the buyer.

- **Orion Electronics** is concerned about maintaining adequate inventory levels. The firm has an **annual cost of goods sold of $12,000,000** and an **average inventory value of $3,000,000**. Determine the **number of weeks of supply** the company currently holds.

- **BrightStar Electronics** is a consumer electronics manufacturer experiencing high inventory holding costs and wants to check whether its inventory is selling at a reasonable pace. The company reports an **annual cost of goods sold of $9,000,000** and an **average inventory of $1,800,000**. **What is the annual inventory turnover?**